

Supplementary Material

Journal Title: IEEE Robotics and Automation Letters
Reference Number: 26-3841v1
Title of Paper: VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuo-motor Skill Distillation
Authors: Anonymous Authors

A RL Teacher Training and Safety Details

A.1 Asymmetric Teacher Observations

Table 1 details the asymmetric actor–critic observation spaces used during teacher-policy training. The critic receives a 133-dimensional privileged simulator state, whereas the actor receives an 87-dimensional observation consisting only of signals available at deployment. We inject noise into the actor observations during training to improve robustness and reduce the sim-to-real gap. The palm and five fingertips constitute the six hand links. A “twist” is the concatenation of three-dimensional linear and angular velocity. Running mean/variance normalization is applied to both inputs and to the value targets.

Table 1: Asymmetric teacher observations.

Signal	Actor	Critic	Definition
Robot joint position q_t	13 (noisy)	13	Actuated arm–hand joint angles (rad)
Robot joint velocity $\dot{q}_t$	13 (noisy)	13	Actuated arm–hand joint velocities (rad/s)
Hand-link position	18 (noisy)	18	Palm and five fingertip positions: 6×3 Cartesian coordinates
Hand-link velocity	18 (noisy)	36	Actor: linear velocity (6×3); critic: spatial twist (6×6)
Palm contact force	–	3	Privileged Cartesian contact-force vector (N)
Measured joint torque	–	19	Privileged torque measurements for 19 instrumented joints (N m)
Object position	3 (noisy)	3	Object Cartesian position
Object orientation	4 (noisy)	4	Object rotation as a unit quaternion
Object velocity	–	6	Privileged object linear and angular velocity (twist)
Goal position	3	3	Cartesian target position for the lift task
Object scale	1	1	Scalar geometry scale supplied by the environment
Object ID	1	1	Scalar identifier for the current object
Previous action a_{t-1}	13	13	Previous 13-dimensional arm–hand command
Total dimension	87	133	

“–” denotes a signal that is unavailable to the actor. Noisy entries are sensor-like estimates; unmarked actor entries are task context or action history. Critic-only entries are used during simulation training and are not available to the deployed policy.

A.2 Termination Conditions

We define four unsafe conditions that serve both as termination criteria during teacher-policy training and as unsafe-event signals during student-policy distillation. We additionally terminate an episode when it reaches the maximum time horizon; however, a timeout is not classified as an unsafe event. All five termination conditions are checked at 60Hz in the environment frame.

- (a) *Harmful collision*: We monitor all arm–table contacts and trigger this termination when the palm or any fingertip is less than 1 cm above the tabletop, or when any arm link contacts the table. An arm–table contact is registered when the ℓ_1 norm of the filtered contact-force vector exceeds 0.1 N.
- (b) *Object escape*: We define a permissible object workspace above the table that covers the intended manipulation region. We trigger object-escape termination as soon as the object center leaves this space or falls below the table.
- (c) *Palm flipped*: The palm should face downward toward the object and table. We trigger this termination when its downward direction differs from world-down by more than 90° , at which point the nominal grasp geometry is inverted.
- (d) *Hand too far*: We use the same permissible workspace as in (b) and trigger this termination when the palm link leaves it, indicating that the hand has moved too far during exploration.
- (e) *Timeout*: We terminate the episode when the time index reaches the maximum episode horizon, $t \geq T_{\max}$.

Expanding the termination rule used in the main paper gives

$$\mathbb{I}_{\text{term}}(s_t, t) = \mathbb{I}_{\text{coll}}(s_t) \vee \mathbb{I}_{\text{obj}}(s_t) \vee \mathbb{I}_{\text{flip}}(s_t) \vee \mathbb{I}_{\text{hf}}(s_t) \vee \mathbb{I}_{\text{timeout}}.$$

To avoid conflicts between simultaneous termination conditions, we assign the event label using the following descending severity priority: harmful collision $\succ$ object escape $\succ$ palm flipped $\succ$ hand too far $\succ$ timeout.

B VLM Prompt and Inference Details

B.1 Visual-Input Storage and Example

Here, we provide representative examples of the selected visual inputs in Fig. 1. These images illustrate observations associated with the two intervention sources considered by our framework: high teacher–student disagreement and low predicted task success. The algorithm block that follows presents the semantic prompt, which combines these visual samples with recent rollout statistics and instructs the VLM to recommend conservative updates to the corresponding intervention thresholds.



Figure 1: Here we present four pictures for two types of intervention sources. Samples (a)–(c) were selected for high teacher–student disagreement, with ℓ_2 values 9.7506, 17.0997, and 51.9710, respectively; (d) was selected for low predicted success (0.2734). All four records have `unsafe=true`.

The algorithm block gives the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

System message:
You tune safety thresholds and return compact JSON only.

User message:

You are a conservative threshold advisor for DexSafeDagger robot distillation. Do not decide intervention directly. Recommend thresholds for the existing gates.
Task: Tune teacher–student disagreement and predicted–success thresholds for dexterous robot distillation.

Existing arbitration:

- intervene if `teacher_student_l2` $\geq$ `l2_threshold`
- or if `predictor_success` $\leq$ `success_threshold`

Return only strict JSON with keys: `l2_threshold`, `success_threshold`, `confidence`, `reason`.
Keep changes conservative. Prefer small threshold shifts unless the recent window shows clear over- or under-intervention.

Base thresholds and clamps:

- `base_l2_threshold`={base_l2_threshold}
- `allowed_l2_range`=[{l2_min}, {l2_max}]
- `base_success_threshold`={base_success_threshold}
- `allowed_success_range`=[{success_min}, {success_max}]

Current advisor thresholds:

- `current_l2_threshold`={current_l2_threshold}
- `current_success_threshold`={current_success_threshold}

Recent rollout statistics:

```
{
  "frame": {frame}, "intervention_rate": {beta}, "step":
{step},
  "sample_count": {N}, "l2_threshold": {tau_delta},
  "success_threshold": {tau_Q}, "unsafe_episode_rate":
{r_unsafe},
  "l2": {"min": {v}, "max": {v}, "mean": {v},
    "p10": {v}, "p50": {v}, "p90": {v}},
  "success": {"min": {v}, "max": {v}, "mean": {v},
    "p10": {v}, "p50": {v}, "p90": {v}},
  "unsafe_reason_prop": { "hand_too_far": {v},
    "harmful_collision": {v}, "object_out_of_bound":
{v},
    "palm_flipped": {v}},
  "visual_buffer_size": {B}, "visual_samples_attached":
{M},
  "visual_samples": [{ "env_id": {v}, "frame": {v},
    "image_attached": true, "image_key": {v},
    "source": {high_l2 | low_success | unsafe_triggered},

    "l2_threshold": {v}, "success_threshold": {v},
    "teacher_student_l2": {v}, "predictor_success":
{v},
    "object_id": {v}, "object_name": {v},
    "object_env_count": {v},
    "step": {v}, "unsafe": {true | false}}, ...]
}
```

For each selected visual sample $i=1, \dots, M$:
Visual sample {i} metadata: {the corresponding record above, excluding "image_attached"}
<IMAGE {i}: JPEG data URL, detail="high">

B.2 Model and Input Configuration

We used OpenAI GPT-5.5 with the rollout statistics above and up to eight high-detail RGB images encoded as JPEG data URLs. We queried the model after 2,000 steps and every 10,000 steps thereafter (10 calls over 100,000 steps). Archived reports from the earlier low-detail setting contain 30,641 prompt and 4,841 completion tokens (35,482 total).

B.3 Output Processing and Threshold Updates

The requested machine-readable output is a top-level JSON object with the following schema:

```
{
  "l2_threshold": <number>, "success_threshold": <number>,
  "confidence": <number>, "reason": <string>
}
```

The two returned thresholds are absolute candidates. Each candidate is clipped to a fixed range around its initial value and then exponentially smoothed with the previously applied threshold. For $x \in \{\delta, Q\}$,

$$\bar{\tau}_x^{(k)} = \text{clip}\left(\hat{\tau}_x^{(k)}, c_x^{\min} \tau_x^{(0)}, c_x^{\max} \tau_x^{(0)}\right), \quad \tau_x^{(k+1)} = (1 - \alpha) \tau_x^{(k)} + \alpha \bar{\tau}_x^{(k)}.$$

We use $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. The updated values are applied from the next control step through the intervention gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q increases teacher intervention. The `confidence` and `reason` fields are logged for diagnostics and do not alter the update. This adaptation is used only during simulation distillation; physical deployment uses the frozen RGB student alone.

If a VLM request fails or its output cannot be parsed, the controller retains the previously applied thresholds and tries again at the next scheduled update. None of the reported trials encountered a condition that activated this fallback.

C Additional Experimental Results

C.1 Bellman Critic Evaluation

We use the resulting student and fixed-threshold Bellman critic from ablation (e) to evaluate the critic’s performance. Both networks are frozen during evaluation, and the critic runs in parallel as a passive evaluator. We evaluate 200 episodes for each of the 40 training objects in simulation. Table 2 reports the observed lift-success and unsafe-episode rates together with the task-outcome prediction accuracy, the unsafe-episode detection rate, and its corresponding false-negative rate.

The lift-success rate and unsafe-episode rate characterize the student policy’s ability to accomplish the task and the frequency of unsafe outcomes, respectively. Task-outcome prediction accuracy is the percentage of episodes for which the critic correctly predicts the task outcome. The unsafe-episode detection rate is the percentage of unsafely terminated episodes that are flagged by the critic before termination. The unsafe false-negative rate is the complementary percentage of unsafe episodes that the critic does not flag before termination.

Table 2: Evaluation of the frozen student and fixed-threshold Bellman critic obtained from ablation (e) under student-controlled rollouts.

Metric	Result
Lift-success rate	86.8
Unsafe-episode rate	11.9
Task-outcome prediction accuracy	90.3
Unsafe-episode detection rate	94.2
Unsafe false-negative rate	5.8

C.2 Ablations Across Random Seeds

We independently train four intervention variants—Vanilla DAgger (b), Vanilla SafeDAgger (c), critic-based intervention (e), and VLM-guided intervention (f)—using five random seeds, $\{40, 41, 42, 43, 44\}$. For each variant and seed, we freeze the resulting student and evaluate it for 200 episodes on each of the 40 training objects in simulation, yielding 8,000 evaluation episodes per seed. Because all variants use the same number of training samples and the teacher-intervention rates exhibit only minor variation across seeds, Table 3 reports only the final lift-success and unsafe-episode rates as the mean $\pm$ standard deviation across the five seeds.

Table 3: Comparison of Vanilla DAGger, Vanilla SafeDAGger, critic-based intervention, and VLM-guided intervention across five random seeds {40, 41, 42, 43, 44}. Lift success and unsafe rate are reported as mean $\pm$ standard deviation across seeds.

Metric	Vanilla DAGger (b)	Vanilla SafeDAGger (c)	Critic-based intervention (e)	VLM-guided intervention (f)
Lift Success (%) $\uparrow$	87.1 $\pm$ 0.99	86.3 $\pm$ 0.92	86.8 $\pm$ 0.74	87.5 $\pm$ 0.98
Unsafe Rate (%) $\downarrow$	17.4 $\pm$ 0.52	16.7 $\pm$ 0.48	11.9 $\pm$ 0.58	9.3 $\pm$ 0.39

C.3 Complete Real-World Evaluation and Demonstrations



Figure 2: **Sixteen-object demonstration inventory:** (1) Horse, (2) Teapot, (3) Sphinx, (4) Dragon, (5) Homer bust, (6) Ogre, (7) Cheburashka, (8) Lucy statue, (9) Beethoven bust, (10) Igea bust, (11) Dog pair, (12) Cow statue, (13) Buddha statue, (14) Blub character, (15) Monkey head, and (16) Dinosaur.

We report results for the remaining 16 real-world test objects that are not shown in the manuscript’s six-object table, using the same success and safety breakdown. The two eight-object blocks below separate task success from the observed unsafe failure modes.

Table 4: Per-object results for the remaining 16 objects (30 episodes per object). **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Chebura shka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	6.7	13.3	13.3	23.3	10.0	6.7	16.7
Harmful collision	6.7	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Object escape	6.7	3.3	6.7	10.0	10.0	6.7	3.3	10.0
Palm flipped	3.3	0.0	3.3	0.0	6.7	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	3.3	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	Dog Pair	Cow statue	Buddha statue	Blub	Monkey head	Dinosaur
Lift Success	86.7	73.3	70.0	63.3	73.3	60.0	66.7	56.7
Unsafe Rate Avg	6.7	13.3	33.3	40.0	33.3	43.3	36.7	46.7
Harmful collision	0.0	3.3	13.3	10.0	13.3	23.3	16.7	26.7
Object escape	6.7	3.3	13.3	16.7	13.3	16.7	16.7	16.7
Palm flipped	0.0	3.3	3.3	10.0	3.3	3.3	3.3	3.3
Hand too far	0.0	3.3	3.3	3.3	3.3	0.0	0.0	0.0

Table 5 summarizes performance across all 22 objects. The mean rates reproduce the pooled results in Main Paper Table IV; the 95% confidence intervals account for the 420 seen-object episodes and 240 unseen-object episodes.

Table 5: Aggregate real-world performance across 14 seen and 8 unseen objects. Mean rates are pooled over 30 episodes per object (420 seen and 240 unseen episodes); parentheses report Wilson 95% confidence intervals.

Metric (%)	Seen mean (95% CI)	Unseen mean (95% CI)
Lift success	83.6 (79.7–86.8)	64.6 (58.3–70.4)
Unsafe rate avg.	14.7 (11.7–18.5)	39.2 (33.2–45.5)
Harmful collision	4.7 (3.1–7.2)	18.8 (14.3–24.2)
Object escape	8.1 (5.9–11.1)	15.8 (11.8–21.0)
Palm flipped	1.4 (0.7–3.1)	3.3 (1.7–6.4)
Hand too far	0.5 (0.1–1.7)	1.3 (0.4–3.6)

C.4 Physical Baseline Comparison

We deploy the frozen RGB student policies resulting from ablations (e) and (f) on the real robot. Ablation (e) uses our RL teacher-training recipe with the stationary DexSafeDagger intervention policy, whereas ablation (f) uses the same teacher recipe with VLM-guided threshold adaptation. We evaluate both policies using the same objects, initial-condition protocol, and number of trials. We report lift-success and unsafe-event rates separately for the 14 seen and 8 unseen objects in Main Paper Table IV(b). The comparison shows that the student trained with VLM-augmented intervention achieves a comparable real-world task-success rate while exhibiting a lower unsafe-event rate for both seen and unseen objects.